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SURVEY ARTICLE

PoseAnalyser: A Survey on Human Pose Estimation

Sakshi Kulkarni[1] · Shubham Deshmukh[1] · Favin Fernandes[1] · Aniket Patil[2] · Vaishali Jabade[1]

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Abstract
Human pose estimation is the process of detecting the body keypoints of a person and can be used to classify different poses. Many researchers have proposed various ways to get a perfect 2D as well as a 3D human pose estimator that could be applied for various types of applications. This paper is a review of all the state-of-the-art architectures based on human pose estimation, the papers referred were based on the types of computer vision and machine learning algorithms, such as feed-forward neural networks, convolutional neural networks (CNN), OpenPose, MediaPipe, and many more. These different approaches are compared on various parameters, like the type of dataset used, the evaluation metric, etc. Different human pose datasets, such as COCO and MPII activity datasets with keypoints, as well as specific application-based datasets, are reviewed in this survey paper. Researchers may use these architectures and datasets in a range of domains, which are also discussed. The paper analyzes several approaches and architectures that can be used as a guide for other researchers to assist them in developing better techniques to achieve high accuracy.

Keywords Convolutional neural networks · Computer vision · Human pose estimation · Machine learning · MediaPipe · OpenPose

Introduction

Many researchers considered human pose estimation to be a broad area of research that had been explored using a range of different approaches. In the initial stages, many researchers started with sensors or hardware implementations to detect the pose of a particular human being. The research implies that we have a better chance of achieving our main objective now since the computer vision domain and machine learning algorithms have evolved.

Human pose estimation refers to the detection of important body joints or keypoints from an image or video of a person's body. Connecting those keypoints to form a joint is also an important step after detecting those keypoints. We can estimate a person's pose from a particular image frame after accurately analyzing the keypoints and body joints and being able to visualize them. This type of method can be used in a wide range of sectors, including sports, medicine, athletics, engineering, and so on.

The evolution of human pose estimation consisting of some traditional methods over the past several years of research will be provided in this research [42]. The introduction of deep neural networks, convolutional neural networks, and the involvement of computer vision resulted in the most significant advances in the human pose estimation area. In our survey-based research, we have considered a variety of approaches to address the problem of human pose estimation to study the earlier research that was conducted using various methodologies. These architectures, which have components of neural networks, aim to produce a framework that can achieve good accuracy in real time while simultaneously being computationally cheap. To evaluate which of these frameworks is most suited for which application, this survey will help to summarize various aspects of all these different

✉ Sakshi Kulkarni
sakshi.nkulkarni@gmail.com

Shubham Deshmukh
shubhamdshmkh1@gmail.com

Favin Fernandes
favin.fernandes@gmail.com

Aniket Patil
aniket.patil@ifm.com

Vaishali Jabade
vaishali.jabade@vit.edu

1 Vishwakarma Institute of Technology, Pune, India

2 Ifm Engineering Private Limited, Pune, India

frameworks. Based on the survey, BlazePose framework is one of the best suited frameworks; the approach and architecture are discussed in the sections below.

Pose Estimation Methods

Pose estimation is the process of locating, identifying, and tracking the number of keypoints or joints on an individual. In humans, these keypoints are major joints such as the elbow, wrists, and knee. The keypoints seem to be in various locations as humans stretch their legs or bend their arms in various ways compared to one another. Two-dimensional (2D) and three-dimensional (3D) images are used to estimate various human poses. In 2D, pose estimation is used to calculate the position of keypoints in 2D space. To recognize 2D human poses in single frames [1], X and Y coordinates are determined for each keypoint. Most approaches train separate 2D pose models before combining them in the correct sequence to understand human motion from video [2, 3]. A 2D image can be transformed into a 3D object for 3D pose estimation by adding a z-dimension to the 2D pose estimation.

Based on the people in the image or video frame, we can differentiate between single-person and multi-person pose estimation [38]. A single-person pose estimation [4, 5] involves only one person being present throughout the video. Multi-person pose estimation is more difficult since the positions and locations of arms or limbs in an image are unknown for each individual person [6]. To solve this problem, the most popular methods used are: (1) top-down approach, which is the easiest approach for human detection; predicting the body parts and calculating the poses for each person; and (2) the bottom-up approach, which involves detecting all human body parts in an image or video and then grouping all body parts that belong to a particular person [7–9].

Different Pose Estimation Approaches

Traditional Method

Traditional methods refer to those algorithms and techniques that do not use the concepts of convolutional neural networks (CNN), recurrent neural networks (RNN), long short-term memory (LSTM) networks, etc. These methods are general machine learning algorithms used for predictive analysis based on vector or cluster principles. In this part, generic pose estimation techniques using machine learning algorithms will be discussed [12].

In 2010, S. Wang, et al. proposed a novel estimation technique for detecting different poses based on the algorithm of the Adaboost technique. Each pixel is classified based on the classifier that is used on an input, i.e., a normalized image that combines the bottom-up joint position classifier and the top-down skeleton classifier. For the training of the classifier, histogram of oriented gradients (HOG) features were extracted from the image and passed. The outcome of this project was the pose estimation of people far from or near the camera [10].

In the paper presented in 2021, the authors suggested a technique to use, i.e., a dimensionality reduction technique to reduce the number of parameters required to train the model using Kmeans, principal component analysis (PCA), and PCA with Kmeans. The techniques reduce the number of features based on input data, the number of principal components for PCA, and the number of clusters for Kmeans. The accuracy of the model was much higher for PCA than for Kmeans [11]. Whereas in the human pose estimation combined with depth information paper, authors used a random forest technique to estimate the keypoints of a person. The detection is based on depth information, after which the depth information is processed. The dataset images are static images rather than video frames [13].

The paper proposed in 2016 by S. Nunez Godoy, et al. uses a sensor-based model to collect data on individuals sitting. The sensors used were strength and ultrasonic sensors. The readings of this data are then passed to PCA for dimensionality reduction and then passed to *k*-nearest neighbors (KNN) for the classification of the pose. By reducing the computational requirements needed for pose estimation by 33%, the authors achieved an accuracy of 75% [14].

Convolutional Neural Network (CNN)

The convolutional neural network has revolutionized the object detection domain in many ways. Due to its good accuracy for tasks like image recognition and object detection, CNN became a key architecture due to its popularity. CNN is still and will remain based on its backbone architecture of human pose estimation.

The authors of the paper, Pose Estimation on 3-Dimensional Models Using ConvNets, used a CNN architecture commonly used for object detection, called VGG16, to detect keypoints from a given human body image. The proposed methodology (see Fig. 1) uses the generator and discriminator approach to reach optimal accuracy. The model was trained on the FLIC dataset [48] and was then compared with other datasets to show that their dataset was the best among the rest. Other parameters of differentiation were the analysis of the GPU's power consumption and keypoint localization through KNN clustering. The main aim of this paper was to build a 3D pose estimation from 2D images to improve the VFX in films [15].

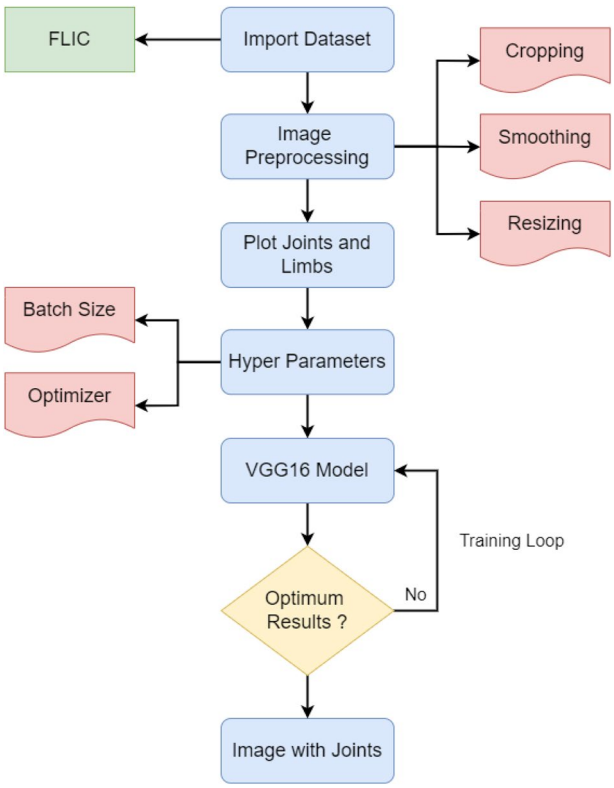

Fig. 1 Proposed flow chart for pose estimation [15]

To estimate different human poses, the tree-based CNN models, non-tree-based CNN models, the proposed hybrid tree-based CNN model, and recent CNN-based HPE models have been compared by the author. The dataset used for this comparison is Leeds sports pose (LSP), containing 2000 images [49]. The pattern classification program (PCP) measure is used for evaluation where the anatomy-based tree-based CNN model with increased nodes achieves the highest accuracy compared to the rest [16]. A literature survey has been conducted in 2019 by A. Singh et al., describing different methods for pose estimation that have been compared using CNN as their backbone. The pose estimation models are DeepCut and multi-person pose detection with localized human joint linkages. These models are compared on the basis of how they achieve their accuracy on the MPII multi-person dataset [17, 47]. The paper [19] might be one of the first attempts to introduce CNN in the human pose estimation domain. The authors of the paper specifically wanted to show that their CNN using the spatial model could outperform the state-of-the-art architecture of that time. The FLIC dataset [48] was used as a comparison parameter between their model, MODEC, and DPM detectors to detect the wrist, elbow, and shoulder joints of a human with the help of precision thresholds.

Another interesting methodology was proposed by H. Tang et al. in 2019, which the author estimated a person's 3D pose with the help of an RGBD camera, which has a color image of a person and also a depth image. The Faster RCNN was used as a human detector in a given frame. Then that part was given to the Resnet50 architecture, part of the CNN family, to extract the body key-points of that person. The MPII [47] and COCO datasets [46] were used for 2D pose estimation, whereas human3.6 was used for 3D pose estimation. The MPJPE metric was used to compare their model and the Microsoft Kinect SDK, where their model outperformed the other. They also used the sparse feature point matching method to align various points in a color and depth image [21].

To perform various tasks in 2019, M. Li et al. combined various architectures like CNN, RNN, LSTM, top-down, and bottom-up approaches using the MSCOCO dataset for training purposes. ResNet50 is used as a feature extractor, whereas long short-term memory (LSTM) pose machines are added between RNN units by imposing pre-trained weights. Long short-term memory (LSTM) was specifically used to detect key-points from image to video and in a multi-person manner. Average precision was the evaluation metric used to compare other models [28].

In the paper [20], the authors are trying something entirely new in every field of the human pose estimation process. The dataset used in this paper is MSCOCO with the annotation, which is unique compared to the rest of the papers as they are using UV fields. This paper compares three architectures: (1) fully convolutional networks (FCN); (2) RCNN; and (3) cascade of mask-RCNN. All these architectures were tested for single-person and multi-person key-point detection. The metrics for evaluation used were area under the curve (AUC), average precision, and intersection over union (IoU). Among all the three architectures, DensePose RCNN Cascade achieved the highest IoU for multi-person detection tasks with the ResNet50 architecture.

Deep Neural Network (DNN)

Deep neural networks are considered the least complex machine learning algorithms. As the advancement in the field of machine learning increased, people started to solve the human pose estimation problem with DNN.

DeepPose is a paper published by Google that is one of the first attempts to introduce DNN in this research domain. The proposed methodology is to use a cascade of DNN-based regressors to achieve high precision keypoint detection. Various experiments were done on FLIC [48] and LSP [49] datasets with the percentage of correct parts (PCP) as the evaluation metric. As a result, DeepPose outperformed on both datasets when compared to other state-of-the-art architectures, with DeepPose on FLIC achieving the highest accuracy [22].

The researchers developed a multi-source deep architecture with non-linear representations for human pose estimation, as proposed by author Jessika et al. 2019. The proposed architecture is a non-linear DNN model with multiple sources as input. It integrates three data sources non-linearly: appearance score, deformation, and appearance mixture type. The model is trained on three different datasets: LPC, PARSE, and UIUC, with PCP as an evaluation metric. Various combinations of parameters were tested, and in the end, their models outperformed the rest of the state-of-the-art architectures [23].

The DNN is used to predict the part affinity field (PAF) and confidence map from a given image. The images are used for key-point detection and the drawing of skeletons. Along with that, the model also uses pre-trained weights of CAFFE for network loading. The model is trained on the Google colab platform with the Tesla K80 as a GPU with the MPII dataset [47]. After training the proposed model [24], it was tested on images and videos. The results show that the model is able to predict and draw key-points around a single person in a given frame, but not multiple people. This limitation might have occurred due to overlapping of predicted keypoints, which could be improved in the future.

Long Short-Term Memory (LSTM)

Long short-term memory (LSTM) generally works well on time series data, since pose estimation activities or human pose tracking are trained on a series of images. Long short-term memory (LSTM) works well to predict poses using the recurrent neural learning concept [25]. Long short-term memory (LSTM) has feedback connections, like traditional feed-forward neural networks.

In this paper [26], the authors proposed Unipose, a model based on the Waterfall atrocity spatial pooling architecture developed by them. The model consists of joint localization along with contextual segmentation to predict the particular human keypoints on the human body without the aid of a statistical post-processing method. The dataset used for the model to train was LSP [49] and MPII [47]. The backbone of the model is based on ResNet and the waterfall model proposed by the authors. Unipose attained good results for estimating the poses of a single person in videos and images based on the efficiency of progressive filtering in the cascade architecture along with stabilizing the pyramid configuration.

For the detection of different poses of multiple people within a single frame, L. Chen, C. Yu, and L. Chen proposed a methodology using long short-term memory (LSTM) architecture. Authors have used deconvolution layers in the single frame architecture on the residual network, i.e., ResNet, to retain image information. The model was trained on COCO [46] and PoseTrack 2018. The model can be run on videos for multi-person pose estimation [27].

The authors of the paper, long short-term memory (LSTM) pose machines [29], have proposed a novel architecture using recurrent neural networks that also includes LSTM to process video frames. Long short-term memory (LSTM) contributed to the efficient usage of temporal information and predicted the results across video frames. The authors have also researched and visualized the memory cells inside the long short-term memory (LSTM) architecture and have detailed the underlying dynamics and processes of the memory during the estimation of poses for different frames in the video. The datasets used for the experiment were the Pen Action and Sub-JHMDB.

OpenPose: Real-Time Multi-person 2D Pose Estimation Using Part Affinity Fields

For multi-person human pose estimation, OpenPose is a commonly used bottom-up method that first detects human body parts or keypoints corresponding to each person in the image, then allocates keypoints to separate people. Using the first few layers of the VGG19 architecture, the OpenPose network extracts features from an image [30, 32]. These features are then sent to two convolutional layer branches (see Table 1), which run in parallel. A first branch predicts confidence maps, each of which shows a different part of the human skeleton. The first branch predicts a number of confidence maps, while the second branch predicts a set of part affinity fields.

A part affinity field [31] is a non-parametric representation that is used to link anatomical structures. It is a set of 2D vector fields and is also used to represent the location and orientation of the limbs. Using an image as input with a group of detected human body parts, K-partite matching implies that the edges can be divided into k-groups of nodes with no connections between limbs for human body part detection. The edges of the graph are weighted according to the part affinities shown in 2017, Cao et al. proposed a series of bipartite graphs by reducing the initial k-partite.

Table 1 OpenPose architecture with two parallel branches

Initial convolution layer	Refinement convolutional layer
Conv 3×3/128	Conv 7×7/128
Conv 3×3/128	Conv 7×7/128
Conv 3×3/128	Conv 7×7/128
Conv 3×3/128	Conv 7×7/128
	Conv 7×7/128
	Conv 1×1/128

BlazePose: On-Device Real-Time Body Pose Tracking

BlazePose is a high-resolution human gesture detection technique that is also known as MediaPipe. The MediaPipe pose keypoint model predicts 33 pose landmarks and can also predict the whole body from RGB video frame segmentation masks representing either person segmentation or background segmentation. In research, Valentin Bazarevsky and associates trained BlazePose Full, which has 3.5 million parameters, and also BlazePose Lite, which has 1.3 million parameters [33]. On the AR dataset, BlazePose performed considerably worse than the OpenPose model. In fitness applications, BlazePose outperforms OpenPose.

ShuffleNet: An Extremely Efficient Convolutional Neural Network for Mobile Devices

ShuffleNet is a very efficient and robust convolutional neural network architecture, developed specifically for mobile devices [35]. The dataset used in the paper is MS COCO object detection [46] with a lower top-1 error. ShuffleNet is a multistage architecture jointly responsible for the detection and association of human body parts. To reduce computation costs with the same accuracy, the new architecture was developed with pointwise group convolution and channel shuffle. Based on the ShuffleNet unit's building blocks (see Fig. 2) [35] and for the detection of 2D multi-person pose estimation, the authors of the paper [34] used the ShuffleNet architecture in a real-time application.

In MobileNetV1, a bottleneck unit with depthwise convolution is used. The first and second 1×1 convolutions of the ShuffleNet unit are replaced by group convolutions. A channel shuffle is applied after the first 1×1 convolution. Afterward, when the stride is applied to the ShuffleNet unit, a 3×3 average pooling is applied. To increase the channel dimension, channel concatenation replaces element wise addition [35].

HRNet

The HRNet provides high-resolution representations throughout the workflow by sequentially adding high-resolution to low-resolution convolution streams prior to actually linking the multiresolution streams in parallel, and the semantic strength of all these high-resolutions to low-resolution representations has expanded at the rate of applying multiresolution connections to increase high-resolution representations using low-resolution representations and inversely. In comparison to ResNet-50 simple baseline, the number of inputs that influence performance in terms of estimation performance, parameter complexity, and computation complexity (GFLOPS) is less. The researchers applied an approach described in [36] with a 256×192 input size that performs the simple baseline with a 384×288 input size. In 2020, Manchen Wang et al. developed a top-down approach for multiple person human pose estimation as well as tracking in videos, which consists of three main parts: (1) a clip tracking network that

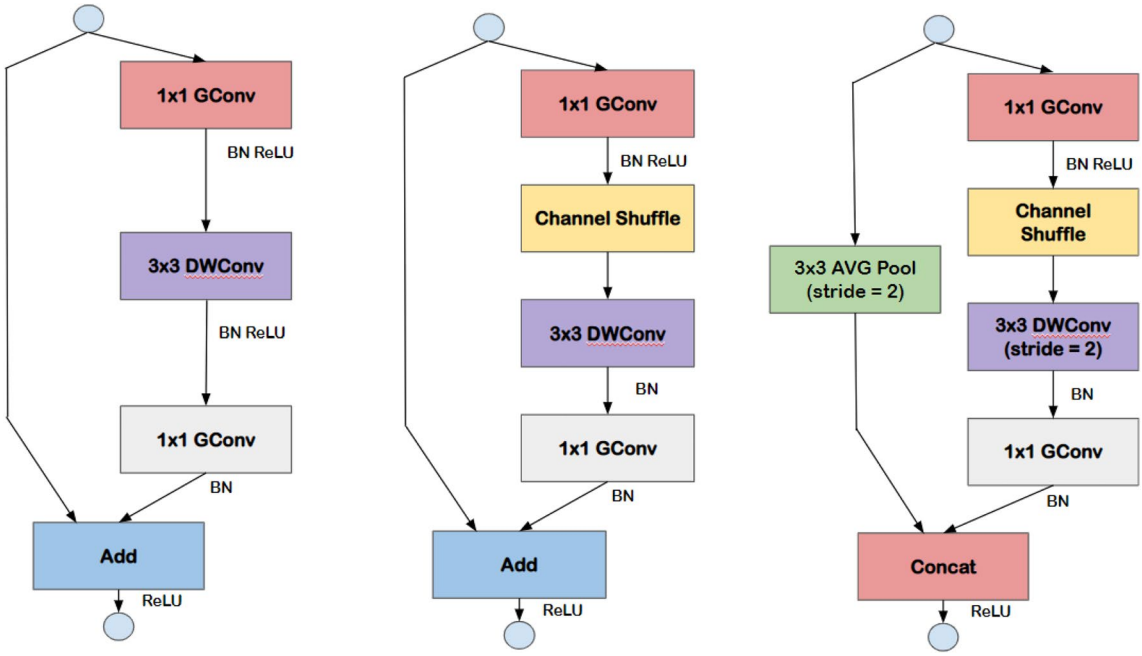

Fig. 2 **a** Bottleneck unit with depthwise convolution, **b** ShuffleNet unit with channel shuffle and pointwise group convolution, **c** ShuffleNet unit with stride = 2 [34, 35]

detects and tracks body joints at the same time; (2) a video tracking that merges the fixed length tracks produced by the clip tracking network into arbitrary length tracks; and (3) based on spatial and temporal smoothing terms, a spatial temporal merging process, one which optimizes joint locations [37].

Other Approaches

Pose recognition is commonly used in a variety of applications, such as the training of human pose estimation using simple preprocessing techniques, including the COCO objects dataset [46], by taking the input images and then creating the bounding box for visualizing crowded scenarios [39]. In addition, human pose estimation in outdoor sports broadcast areas constitutes approximately 70% of the outdoor sports footage being used to analyze 3D images of human body actions [40].

The MobileNetV2 backbone architecture has been used to estimate 3D human poses by S. Choi et al. The reason for choosing MobileNetV2 was to run the model efficiently on a mobile phone and achieve good accuracy. The authors even improved MobileNetV2 by adding skip concatenation and parametric activation functions. After training, a comparison with state-of-the-art architecture was made, taking into account all the factors, and their model was determined to be the winner [18].

The authors, Sha et al., collected several test images depending on the various conditions and also tested the proposed methodology in other postures such as standing and sitting. To analyze the overall performance and improve the pose estimation process, the method was used in combination with the depth information captured by the Kinect as well as the preprocessing of depth images [13]. In 2019, other researchers used the LCR-Net architecture [41], which consists of three major components: (1) a dataset of human body poses in various locations for both single- and multi-person detection of the MPII 2D pose benchmark in real time [47]; (2) a classification algorithm with different scores for different pose proposals; and (3) a regression model that refines pose proposals for both 2D and 3D images.

In 2017, Vasileios Belagiannis and Andrew Zisserman proposed [45] a ConvNet model with an architecture that combines a feed-forward module, which runs only once, with a recurrent module that can be operated multiple times. The model was then trained and improved performance with auxiliary losses incorporated and analyzed to analyze the keypoint visibility to predict 2D human body poses.

Dataset

Many human pose estimation research papers mainly use the COCO [46] and MPII [47] datasets, and some research papers are based on LSP, PoseTrack 2017, PoseTrack 2018, and FLIC datasets for particular applications-based research. In the paper [33], the BlazePose architecture was trained by taking input data images from BlazeFace, BlazePalm, and the AR dataset, as well as the COCO dataset [46]. The AR face dataset provides 4,000 color images of 70 men and 56 women. Faces with various facial emotions are featured in frontal views.

Common Objects in Context (COCO) Dataset

The COCO dataset was created to represent a wide range of objects that we encounter in our everyday routine. It is a labeled database providing information for training-supervised computer vision models that can recognize the objects in the dataset [46]. The model can be fine-tuned to learn various other tasks with a custom dataset after it has been trained on the COCO dataset. There are different dataset types in the COCO dataset, such as (1) object detection, where objects are annotated with a bounding box and each class is labeled with a specific name; (2) keypoint detection dataset having human body keypoints; and (3) semantic segmentation, where the boundary of objects is labeled with a mask and object classes are labeled with a class label. There are 80 major classes in the COCO dataset.

Osokin proposed a method that achieved first place in the inaugural COCO 2016 keypoints challenge and outperforms the previous results on the MPII multi-person dataset in terms of both performance and reliability [31, 47]. A few researchers [20, 28, 35, 39] also use the MS COCO dataset to detect various human activities, which is a huge object detection, segmented, and labeled dataset published by Microsoft.

MPII Human Pose Dataset

The MPII dataset contains over 25,000 images, including people with annotated body joints and about 40,000 images of single people as well as multiple people in one frame. The images were gathered in a structured manner using a classification of everyday human activities. The data comprise 410 different human activities, with labels associated with each image. Every image was captured from a video on YouTube [47]. Several test images were collected in the paper [13], involving clothing colors, dresses, pants, and also human activities such as walking, standing, dancing, and sitting. The MPII human pose dataset contains 16 body joints,

covering shoulders, elbows, wrists, knees, hips, ankles, and much more. This dataset has used both 2D as well as 3D human pose estimation approaches [30, 41].

The Frames Labeled in Cinema (FLIC) Dataset

The FLIC dataset was obtained from 30 popular Hollywood films and then annotated by Amazon Mechanical Turk, a crowdsourcing marketplace (AMT) [48]. The dataset contains approximately 5000 images and is divided into 80% of training images and 20% of test images, with 11 upper body joints such as the eyes, nose, elbows, shoulders, and wrists [15]. The authors of papers [19] and [22] compared the Leeds sports pose dataset with other existing datasets to get a good evaluation matrix.

Leeds Sport Pose (LSP) Dataset

Leeds sports pose dataset [49] is a collection of images that are used to solve pose estimation problems in sports activities. This dataset contains 2000 annotated images of sportspeople in various poses, collected from Flickr. The dataset can be observed in the JPEG format of each input image. The images have been scaled so that the most recognizable individual seems to be about 150 pixels, with each image having 14 keypoints. From a person-centric point of view, the left and right joints are continuously labeled. In their paper [26], B. Artacho and A. Savaki used the LSP dataset [49] along with the MPII [47] human pose dataset to detect a single person in a video sequence.

Applications of Human Pose Estimation

Thai Sign Language Recognition: An Application of Deep Neural Network

In the paper proposed by Chen et al., the authors are trying to classify various words and numbers in the Thai sign language using the MediaPipe architecture to detect the keypoints of the palm. The dataset used for this research is a set of 100 videos for each word, where 60 of these will be used for training and 40 for testing. These videos have been used to train the model to classify different words using the MediaPipe framework. The MediaPipe framework will detect the 21 key-points of each hand. These 42 keypoints from each image frame of the video will be stored in a CSV file [43].

For the classification of words, they have trained the MediaPipe CSV result with the help of three different architectures related to RNN: (1) LSTM, (2) bi-LSTM, and (3) GRU. The results were compared after training the model in all three different architectures and later hypertuning some parameters to improve accuracy. The accuracy and loss were the parameters taken for comparison where long short-term memory (LSTM) achieved the highest accuracy of 97% and both Bi-LSTM and GRU achieved an accuracy of 94% on the test data. These pre-trained were later used to make a desktop application as well as a mobile application where the long short-term memory (LSTM) model was used on the desktop, whereas the GRU model was used on the mobile as it has limited processing capacity. Even if the model has some limitations, the MediaPipe framework can be a useful tool to extract keypoints and can develop more in the future.

Fall Detection System Based on Real-Time Pose Estimation and SVM

This paper tries to solve the problem of elderly people falling asleep [44]. The authors are trying to detect if an elderly person is falling with the help of pose estimation models and machine learning algorithms in real time through surveillance cameras. Later on, two types of datasets will be used, which will be the basis of comparison: (1) URFD, which mostly contains images of daily human activities, with 42% of falling images taken from two cameras. (2) MCFD, which are videos of people falling and doing normal things captured by eight web cameras.

The proposed methodology is to first detect a human body using the YOLOv5 architecture, and then all the body key-points of the person using the OpenPose framework, giving 14 detected body key-points to SVM and ResNet32 classifiers to detect whether the person is falling or not. For detection purposes, they have kept the camera in a top-down position, which helps detect a fall without being blocked by unwanted objects. The parameters used for evaluation were sensitivity and specificity. After the training process according to the methodology, it was observed that the dataset MCFD performs better than the URFD dataset. In conclusion, an image dataset from multiple angles can be useful for better accuracy.

Summary

In this section, the pose estimation techniques are compared based on certain parameters. This paper surveyed all of the architectures used in human pose estimation, as well as some of the most interesting studies that have been conducted. All the methods have been discussed in brief in the above sections, covering all aspects of the network used along with the approach, estimation type, and dataset used. In most cases, researchers chose a top-down technique over a bottom-up method. However, some architectures combined both approaches (see Table 2).

Table 2 Summary of different pose estimation methods used by different researchers

Approaches	Backbone architecture	Single/multi-person	Top-down/bottom-up	Dataset used	Evaluation metric
Learning human pose estimation features with convolutional networks (2014)	CNN	Single-person	Top-down	FLIC	Precision threshold
DeepPose: human pose estimation via deep neural networks (2014)	DNN	Single-person	Bottom-up	FLIC and LSP	0.69 PCP (percentage of correct parts)
Multi-source deep learning for human pose estimation (2014)	DNN	Single-person	Bottom-up	LPC, PARSE and UIUC	71.0 PCP
Real-time multi-person 2D pose estimation using part affinity fields (2017)	VGG	Multi-person	Bottom-up	COCO and MPII	AP, mAP, percentage of Correct keypoints (PCK)
ShuffleNet: an extremely efficient convolutional neural network for mobile devices (2017)	CNN	Multi-person	Bottom-up	MSCOCO	524 MFLOP
Recurrent human pose estimation (2017)	CNN	Single-person	Top-down	MSCOCO and MPII	Accuracy—85%
Human pose estimation using convolutional neural networks (2017)	CNN	Multi-person	Bottom-up	MPII	Accuracy—87.9%
Cascaded pyramid network for multi-person pose estimation (2018)	ResNet, FPN and Mask RCNN	Multi-person	Top-down	COCO	Average precision (AP), OKS
LSTM pose machines (2018)	RNN and LSTM	Single-person	Top-down	Pen action and sub—JHMDB	RPM (93.5)
Simple baselines for human pose estimation and tracking (2018)	ResNet	Multi-person	Top-down	COCO	AP, mAP
DensePose: dense human pose estimation in the wild (2018)	FCN, R-CNN, mask R-CNN	Multi-person	Bottom-up	MSCOCO	AP, AUC, IoU (0.81)
HRNet: deep high-resolution representation learning for human pose estimation (2019)	ResNet	Multi-person	Bottom-up	COCO and MPII	AP, mAP
Research on 3D human pose estimation using RGBD camera (2019)	Faster R-CNN, RestNet50	Multi-person	Top-down	MPII and COCO	Percentage of correct keypoints (PCK), mean per joint position error (MPJPE)
Human pose estimation for real-world crowded scenarios (2019)	ResNet	Multi-person	Top-down	CrowdPose, JTA	AP, OKS
LCR-Net++: multi-person 2D and 3D pose detection in natural images (2019)	ResNet50, VGG16	Both	Top-down, bottom-up	MPII and Human3.6 M	IoU, PCK, mAP
Multi-person pose estimation using bounding box constraint and LSTM (2019)	CNN, RNN, ResNet50 and LSTM	Multi-person	Bottom-up, top-down	MSCOCO	66.03 AP

Table 2 (continued)

Approaches	Backbone architecture	Single/multi-person	Top-down/bottom-up	Dataset used	Evaluation metric
A study on part affinity fields implementation for human pose estimation with deep neural network (2019)	DNN	Single-person	Bottom-up	MPII	–
OpenPose: Real-time multi-person 2D pose estimation using part affinity Fields (2019)	VGG19	Multi-person	Bottom-up	COCO and MPII	AP, OKS
A multi-person pose estimation with LSTM for video stream (2019)	ResNet with deconvolution	Multi-person	Top-down	PoseTrack 2018, COCO	AP and AR
Pose estimation on 3D nodels using ConvNets (2020)	VGG16	Single-person	Bottom-up	FLIC	–
BlazePose: on-device real-time body pose tracking (2020)	CNN	Single-person	Bottom-up	BlazeFace, BlazePalm, AR and COCO	Intersection over union (IoU)
Combining detection and tracking for human pose estimation in videos (2020)	ResNet	Multi-person	Top-down	PoseTrack2017 and 2018	83.8 mAP (Detection) and 71.6 MOTA (tracking)
A hybrid neural network for graph-based human pose estimation from 2D images (2020)	CNN	Single-person	Bottom-up	LPC	Accuracy—81.1%
UniPose: unified human pose estimation in single images and videos (2020)	ResNet and Waterfall	Single-person	Top-down	LSP and MPII	72.8% PCP (LSP) and 92.7% PCP (MPII)
MobileHumanPose: toward real-time 3D human pose estimation in mobile devices (2021)	MobileNet V2	Multi-person	Top-down	Human3.6 M and MPII	79.6 mm (MPJPE)
Combined top-down/bottom-up human articulated pose estimation using AdaBoost learning (2010)	HOG and Adaboost	Single-person	Bottom-up and top-down	Custom pedestrian dataset	Accuracy—99.9% (125 ms)

Conclusion

This research provides a summary of the methodologies proposed by researchers based on several different parameters and different datasets. Human pose estimation has always been a difficult task to execute, even with the advancements in computer vision and machine learning fields. This paper also reviews multiple approaches that are often used to determine whether the model is single person or multi person, whether the pose detector is 2D or 3D, and whether the bottom-up or top-down strategy should be used. The different types of architecture used to accomplish the detection and the various datasets used to train it were then discussed. A perfect combination of all these parameters will result in an efficient human pose estimation model. Among all these architectures, the BlazePose framework is one of the best suited and commonly used for human pose estimation as it has better accuracy in real-time applications and has limited processing capacity, which can run on almost any electronic device. This model is representative of overcoming limitations as it is the product of many iterations of deep neural networks, convolutional neural networks, and LSTM architectures which had limitations like poor accuracy, high processing time, single-person keypoint detector, and complex architectures. All the prominent architectures up-till now are reviewed in this paper considering all these parameters, which provides a better picture for comparison. This paper could be helpful to those who want to innovate in this field by researching the limitations of these architectures.

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Data Availability The datasets analysed during the current study are included in this published article. For eg. Section "Dataset" and Table 2 consist of relevant information.

Declarations

Conflict of Interest The authors declare no conflict of interest.

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